Joseph Mesches

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Electrical Engineering student at CSU and critical-care flight paramedic focused on machine learning, signal processing, and embedded systems. Builds interpretable ML and DSP pipelines for biomedical and sensor data, from feature extraction through edge deployment.

Education

Bachelor of Science, Electrical Engineering — Expected May 2027
Colorado State University — Major GPA: 3.7 | Cumulative: 3.4 | Dean's List | HKN
Coursework: Digital Signal Processing, Biomedical Signal & Image Processing, Linear Systems, Communications, Microcontrollers & IoT, Embedded Systems / ML

Associate of Applied Science, Paramedicine — December 2019
Arapahoe Community College

Technical Skills

Machine Learning & DSP: classification, ensemble learning, explainable AI, 1D CNNs, feature extraction, adaptive filtering, FFT & windowing, edge inference (Coral TPU), RAG
Programming: Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch, Keras), C/C++, embedded C, MATLAB, SQL, Bash
Tooling: Git, Linux systems administration, Docker, Jupyter, Grafana; KiCad, Altium Designer, LTSpice
Protocols & Hardware: SPI, I2C, UART, CAN; oscilloscopes, logic analyzers, FPGAs

Experience

Student Research Assistant, Radar Systems — Aug 2026 – Present
Prof. Chandrasekar's Radar Lab, Colorado State University
– Developing AQPI Sentinel, a radar-product monitoring interface, and integrating it into the lab's backend systems.

Electrical Engineering Intern — June 2026 – Present
The RMH Group, Inc. (MEP)
– Reverse-engineered an automatic racking robot and conveyor system into AutoCAD drawings and contributed to control-system projects improving plant efficiency; invited to continue through the academic year.

Projects

OpenOrbit Radar Payload Team (Senior Design) — Aug 2026 – May 2027
Project Manager, Colorado State University
– Lead the team building a drone-based synthetic aperture radar (SAR) imaging system; manage schedule, budget, funding, and sponsor relations while contributing to the engineering.

Explainable ECG Clinical Decision-Support System — Spring 2026
ECE 431, Colorado State University
– Led integration on a 4-person team building a five-pipeline ML ensemble (XGBoost + 1D-ResNet CNN over engineered ECG signal-processing features) that classifies 12-lead ECGs into five diagnostic classes at 0.93 macro-AUC, on par with published deep-learning benchmarks.
– Authored the shared DSP library (filtering, R-peak detection, fiducial delineation) and mapped predictions to named clinical criteria for auditable, clinician-reviewable output.

Edge ML Wildlife Tracker — 2024 – Present
Personal
– Built low-latency on-device detection (IP cameras + Coral Edge TPU) with mobile alerts; integrated vision-language models for behavioral classification and pattern-of-life logging.

Leadership

IEEE Student Branch: President (2025–2026), Vice President (2026–Present) — 2025 – Present
Colorado State University
– Lead branch operations, outreach, and communication between students, faculty, and industry partners.

EMS Field Training Officer, Instructor & QA Reviewer — 2015 – 2022
Multiple Colorado EMS Agencies
– Led field-training teams of up to 30 and trained 200+ paramedics; served on quality-assurance and education committees authoring SOPs.

Clinical Experience

Critical Care / Flight Paramedic — 2017 – 2022
UCHealth LifeLine, Air Methods Corporation, Northglenn Ambulance
– Interpreted thousands of 12-lead ECGs and managed advanced monitoring, ventilation, and hemodynamic support for the most critically ill patients across scene, ED, and inter-ICU transport.

Publications, Honors & Certifications

Publication: "Looking Past Dementia Reveals Hidden Life Threats," *Journal of Emergency Medical Services* (2018)
Certifications: Certified Flight Paramedic (FP-C, ICAPP), Critical Care Endorsement (P-CC), Nationally Registered Paramedic
Honors: Walker-Weinshenker Family Scholar (2024–2026), Dean's List, IEEE-HKN